

Introduction to the Practice of
STATISTICS NINTH
EDITION

Moore / McCabe / Craig

Chapter 1:

Looking at Data — Distributions

Lecture Presentation Slides

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Chapter 1

Looking at Data— Distributions

Introduction

1.1 Data

1.2 Displaying Distributions with Graphs

1.3 Describing Distributions with Numbers

1.4 Density Curves and Normal Distributions

1.1 Data

- Cases, labels, variables, and values
- Categorical and quantitative variables
- Key characteristics of a data set

Cases, Labels, Variables, and Values

- ✓ **Cases** are the objects described by a set of data. Cases may be customers, companies, subjects in a study, units in an experiment, or other objects.
- ✓ A **label** is a special variable used in some data sets to distinguish the different cases.
- ✓ A **variable** is a special characteristic of a case.
- ✓ Different cases can have different **values** of a variable.

Label ? Variables? Case? Value?

Study ID	ONSAGE	AGE_med	RACE	GENDER	DUD	MSA	Diabetes	Fatty_liver	Calcification	Lipodystrophy
10000196	4.383562	4.63287671	0	0	2.989733	0	0	0	0	0
10000198	6.041096	7.75342466	0	0	20.53388	3	0	0	0	0
10000201	7.041096	7.34246575	2	0	3.613963	1	0	0	0	0
10000213	20.39452	21.2054795	0	0	9.724846	1	0	0	0	0
10000214	2.282192	2.65479452	0	1	4.468172	0	0	0	0	1
10000215	9.99726	10.0931507	2	0	1.149897	0	0	0	1	0
10000216	4.526027	5.25753425	0	0	8.772074	3	0	0	0	0
10000228	11.17534	11.2630137	0	1	1.051335	0	0	0	0	1
10000256	2.246575	2.75068493	0	0	6.045175	3	0	0	0	1
10000263	2.668493	3.52876712	0	1	10.31622	0	0	0	0	1
10000265	2.520548	3.71506849	0	0	14.32444	0	0	0	2	1
10000267	7.542466	7.70410959	0	0	1.938398	0	0	0	0	0
10000268	7.690411	8.23561644	0	0	6.537988	3	0	0	0	1
10000275	2.473973	2.64931507	0	0	2.102669	3	0	0	0	0
10000296	4.30411	5.40547945	2	1	13.20739	3	0	0	0	1
10000298	8.547945	9.65205479	0	0	13.24025	3	0	0	0	1
10000302	8.430137	8.53424658	0	0	1.24846	0	0	0	0	0
10000304	5.950685	6.12328767	2	1	2.069815	1	0	0	0	0
10000311	4.115068	4.55342466	3	0	5.256674	3	0	0	1	0
10000312	13.36986	13.5369863	0	0	2.004107	0	0	0	0	0
10000314	1.383562	3.07123288	2	0	20.23819	3	0	0	0	0

MSA: myositis-specific autoantibodies

Categorical and Quantitative Variables

- ❑ A **categorical** variable places each case into one of several groups, or categories.
- ❑ A **quantitative** variable takes numerical values for which arithmetic operations such as adding and averaging make sense.

Key Characteristics of a Data Set

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Every data set is accompanied by important background information. In a statistical study, **always ask the following questions:**

- **Who?** What **cases** do the data describe? **How many** cases does a data set have?
- **What?** How many **variables** does the data set have? What are the **exact definitions** of these variables? What are the **units of measurement** for each quantitative variable?
- **Why?** What **purpose** do the data have? Do the data contain the information needed to answer the questions of interest?

CLINIC ID	$\Delta\Delta C_T$	GenStatus	Tissue	DAS-S	DAS-M	DAS-T	Age	Gender	Race	DUDmo	TNFa
10000475	-3.649357557	JDM	PBMC	5	5	10	8.216290212	F	W	15.80287474	GG
10000499	-3.457065582	JDM	PBMC	2	8	10	6.551676934	F	W	5.519507187	GG
10000518	0.178495407	JDM	PBMC	5	1	6	11.1430527	F	W	73.33059548	GG
10001466	-2.878487825	JDM	PBMC	5	9	14	14.90759754	F	W	4.960985626	GG
10000497		JDM	PBMC	7	6.5	13.5	4.73100616	F	W	8.213552361	GA
10000496	1.238426208	JDM	PBMC	5	8	13	3.485284052	M	W	7.12936345	GG
10000522	-2.691696882	JDM	PBMC	nd	nd	nd	2.669404517	M	W	26.64476386	GA
113	-0.807250977	JDM	PBMC	0	0	0	10.95140315	F	W	12.91170431	GG
10000499	-2.091273069	JDM	PBMC	0	0	0	9.327857632	F	W	5.519507187	GG
	-2.041378021	Con	PBMC				10	M	W	N/A	control
	2.687274933	Con	PBMC				9.297741273	M	W	N/A	control
	-1.688948274	Con	PBMC				20.92539357	F	W	N/A	control
	0.146751404	Con	PBMC				17.0403833	F	H	N/A	control
		Con	PBMC				13.65913758	F	W	N/A	control
	-0.980901062	Con	PBMC				14.6036961	F	H	N/A	control
	-0.484334946	Con	PBMC				18.49418207	M	W	N/A	control

1.2 Displaying Distributions with Graphs

- Exploratory Data Analysis
- Graphs for categorical variables
 - Bar graphs
 - Pie charts
- Graphs for quantitative variables
 - Histograms
 - Stemplots

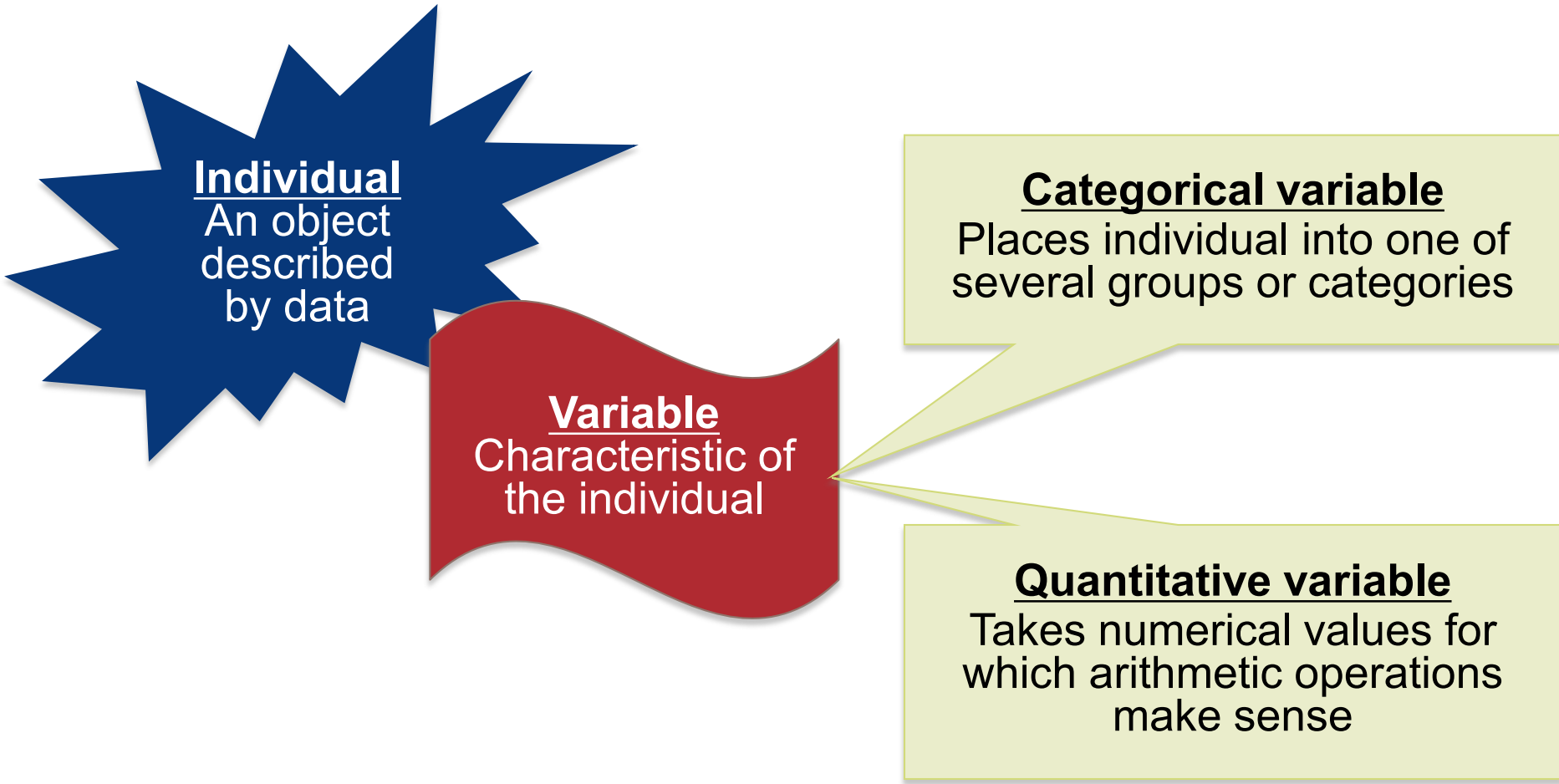
Exploring Data

- Begin by examining each variable by itself (**univariate**). Then move on to study the relationships among the variables.
- Begin with a graph or graphs. Then add **numerical summaries** of specific aspects of the data.

Variables

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For each case in a data set, we record information about characteristics that we call **variables**.



Individual

An object described by data

Variable
Characteristic of the individual

Categorical variable

Places individual into one of several groups or categories

Quantitative variable

Takes numerical values for which arithmetic operations make sense

Categorical or quantitative variables?

Study ID	ONSAGE	AGE_med	RACE	GENDER	DUD	MSA	Diabetes	Fatty_liver	Calcification	Lipodystrophy
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MSA: myositis-specific autoantibodies

Distribution of a Variable

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To examine a single variable, we graphically display its **distribution**.

- The distribution of a variable tells us what values it takes and how often it takes these values.
- Distributions can be displayed using a variety of graphical tools. The proper choice of graph depends on the nature of the variable.

Categorical variable

Pie chart
Bar graph

Quantitative variable

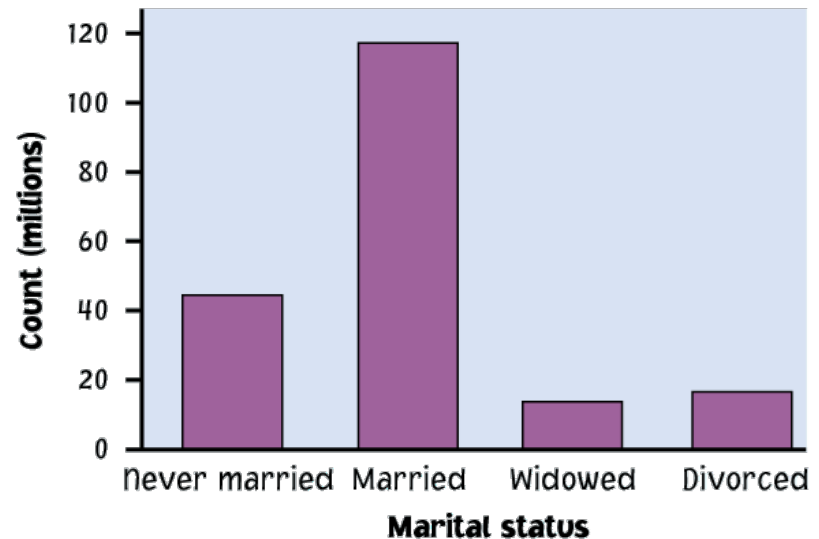
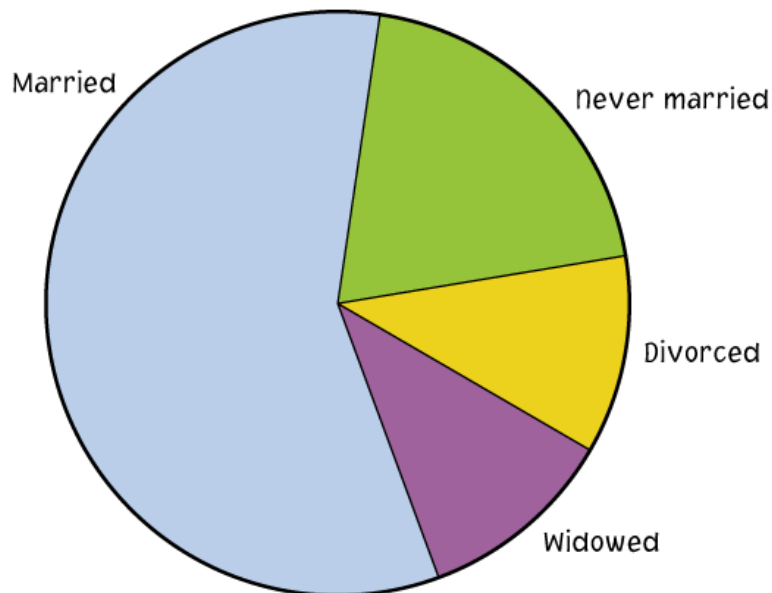
Histogram
Stemplot

Categorical Variables

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The **distribution of a categorical variable** lists the categories and gives the **count** or **percent** of individuals who fall into each category.

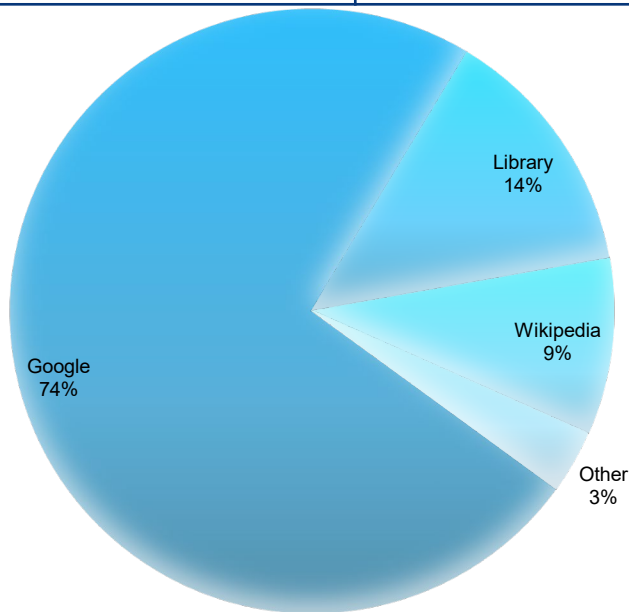
- **Pie charts** show the distribution of a categorical variable as a “pie” whose **slices are sized** by the counts or percents for the categories.
- **Bar graphs** represent categories as bars whose **heights** show the category counts or percents.



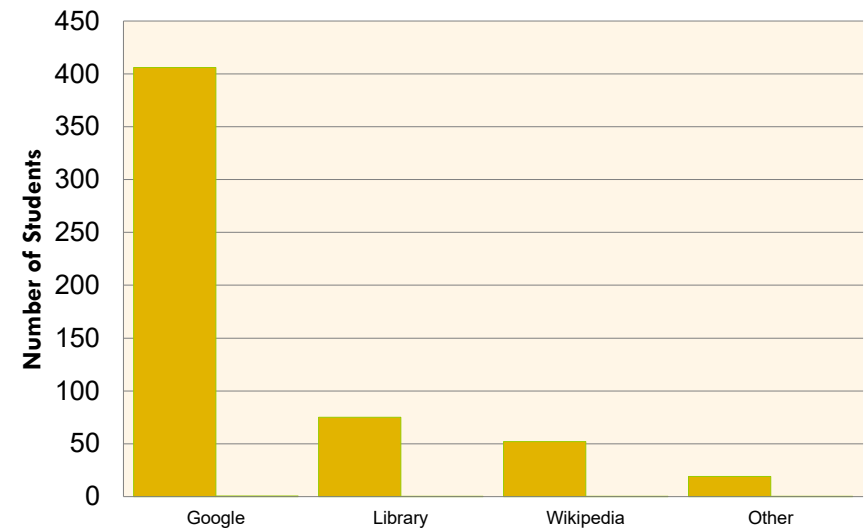
Pie Charts and Bar Graphs

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Resource	Count	Percent of total
Google or Google Scholar	406	73.6%
Library database or website	75	13.6%
Wikipedia or online encyclopedia	52	9.4%
Other	19	3.4%
Total	552	100.0%



Online Research Sources



Quantitative Variables

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The **distribution of a quantitative variable** tells us what values the variable takes on and how often it takes those values.

- **Stemplots** separate each observation into a stem and a leaf that are then plotted to display the distribution while maintaining the original values of the variable.
- **Histograms** show the distribution of a quantitative variable by using bars. The height of a bar represents the number of individuals whose values fall within the corresponding class (or interval).

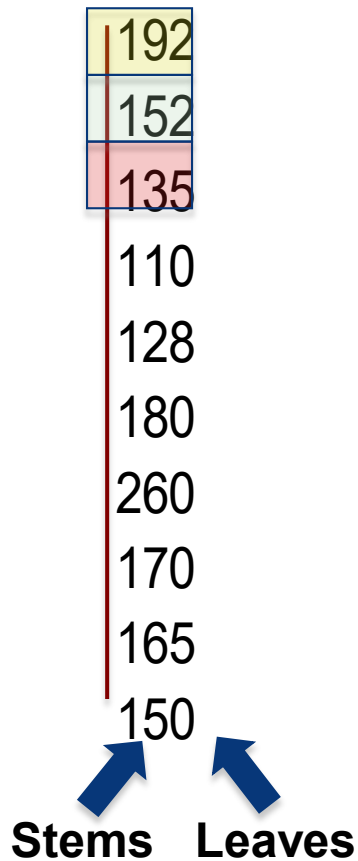
To construct a stemplot:

- Separate each observation into a **stem** (all but the rightmost digit) and a **leaf** (the remaining digit).
- Write the stems in a vertical column; draw a vertical line to the right of the stems.
- Write each leaf in the row to the right of its stem; order leaves if desired.

Stemplots 2

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Example: Weight Data — Introductory Statistics Class



110
120
185
165
212
119
165
210
186
100

195
170
120
185
175
203
185
123
139
106

10
11
12
13 5
14
15 2
16
17
18
19 2
20
21
22
23
24
25
26

Key

20|3 means
203 pounds

Stems = 10s
Leaves = 1s

Stemplots 3

If there are **very few stems** (when the data cover only a very small range of values), then we may want to create more stems by **splitting** the original stems.

Example: If all of the data values are between 150 and 179, then we may choose to use the following stems:

15		
15		Leaves 0–4 would go on each upper stem (first
16		“15”), and leaves 5–9 would go on each lower
16		stem (second “15”).
17		
17		

Histograms 1

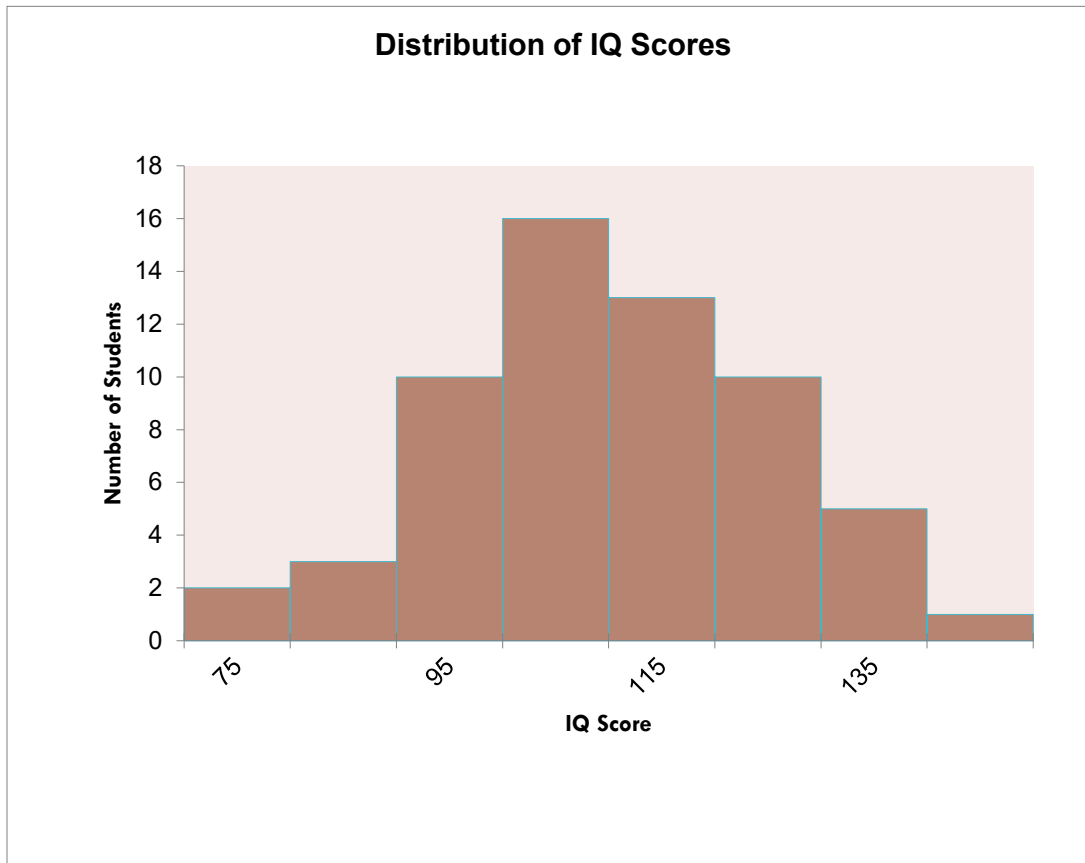
For **large datasets** and/or quantitative variables that take many values:

- Divide the possible values into **classes** or **intervals of equal widths**.
- Count how many observations fall into each interval. Instead of counts, one may also use percents.
- Draw a picture representing the distribution—each bar height is equal to the number (or percent) of observations in its interval.

Histograms 2

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Example: IQ Scores – 60 5th graders



Class	Count
$75 \leq \text{IQ Score} < 85$	2
$85 \leq \text{IQ Score} < 95$	3
$95 \leq \text{IQ Score} < 105$	10
$105 \leq \text{IQ Score} < 115$	16
$115 \leq \text{IQ Score} < 125$	13
$125 \leq \text{IQ Score} < 135$	10
$135 \leq \text{IQ Score} < 145$	5
$145 \leq \text{IQ Score} < 155$	1

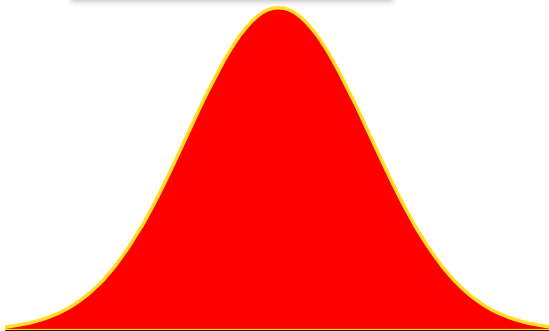
Examining Distributions 1

- In any graph of data, look for the **overall pattern** and for striking **deviations** from that pattern.
- You can describe the overall pattern by its **shape**, **center**, and **spread**.
- An important kind of deviation is an **outlier**, an individual that falls outside the overall pattern.

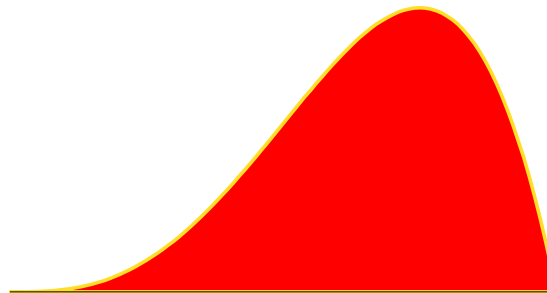
Examining Distributions 2

- A distribution is **symmetric** if the right and left sides of the graph are approximately mirror images of each other.
- A distribution is **skewed to the right** (right-skewed) if its right tail is substantially longer than its left tail.
- A distribution is **skewed to the left** (left-skewed) if its left tail is substantially longer than its right tail.

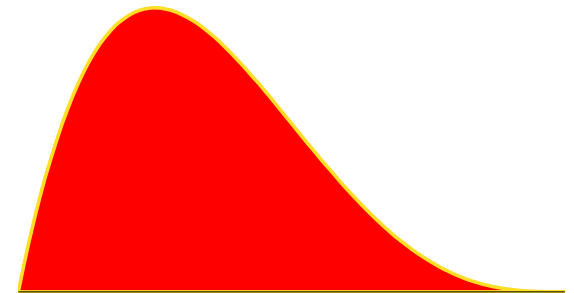
Symmetric



Left-skewed

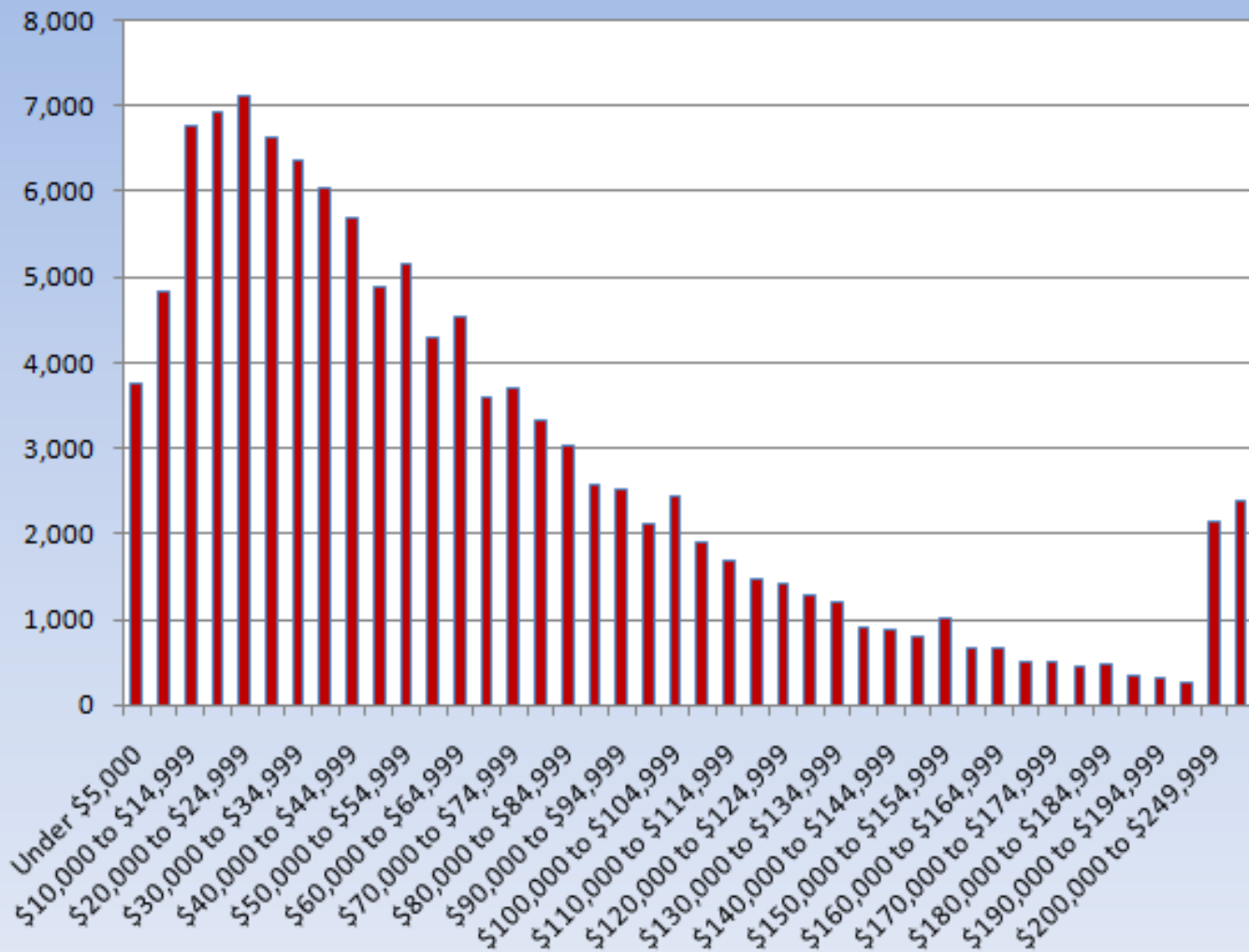


Right-skewed

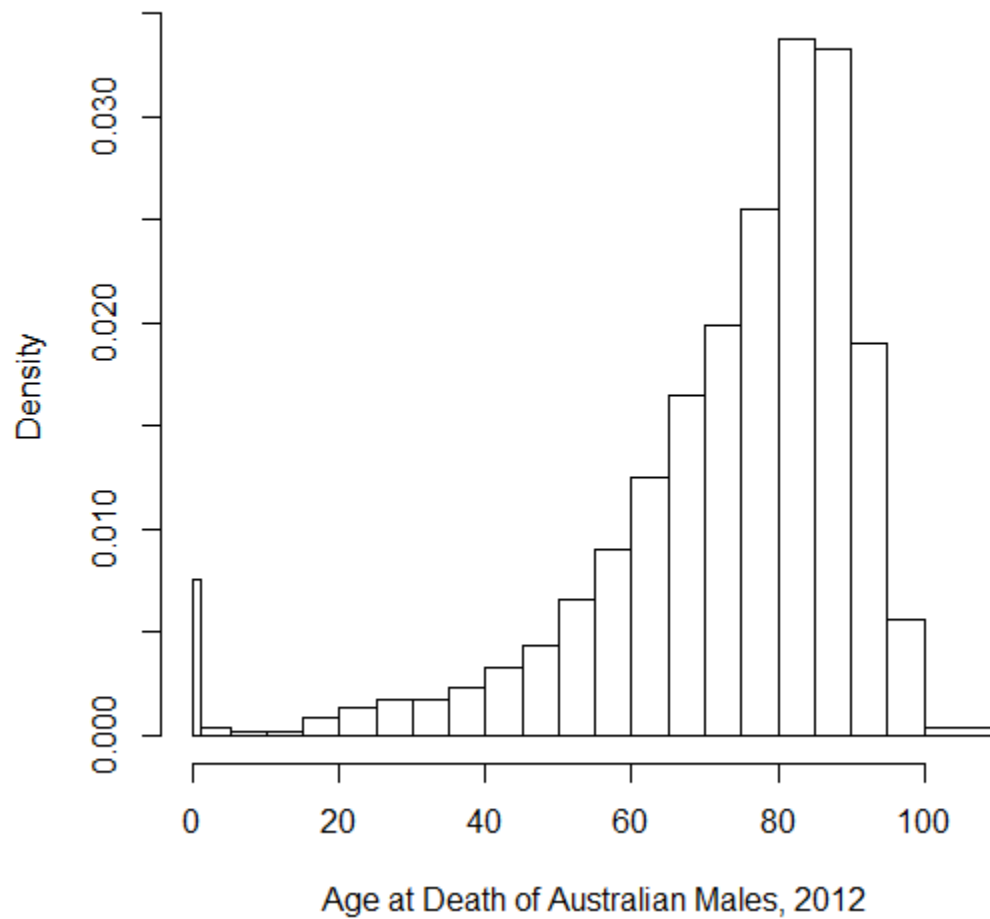


U.S. Household income

www.doctorhousingbubble.com



Histogram of Age at Death of Australian Males, 2012

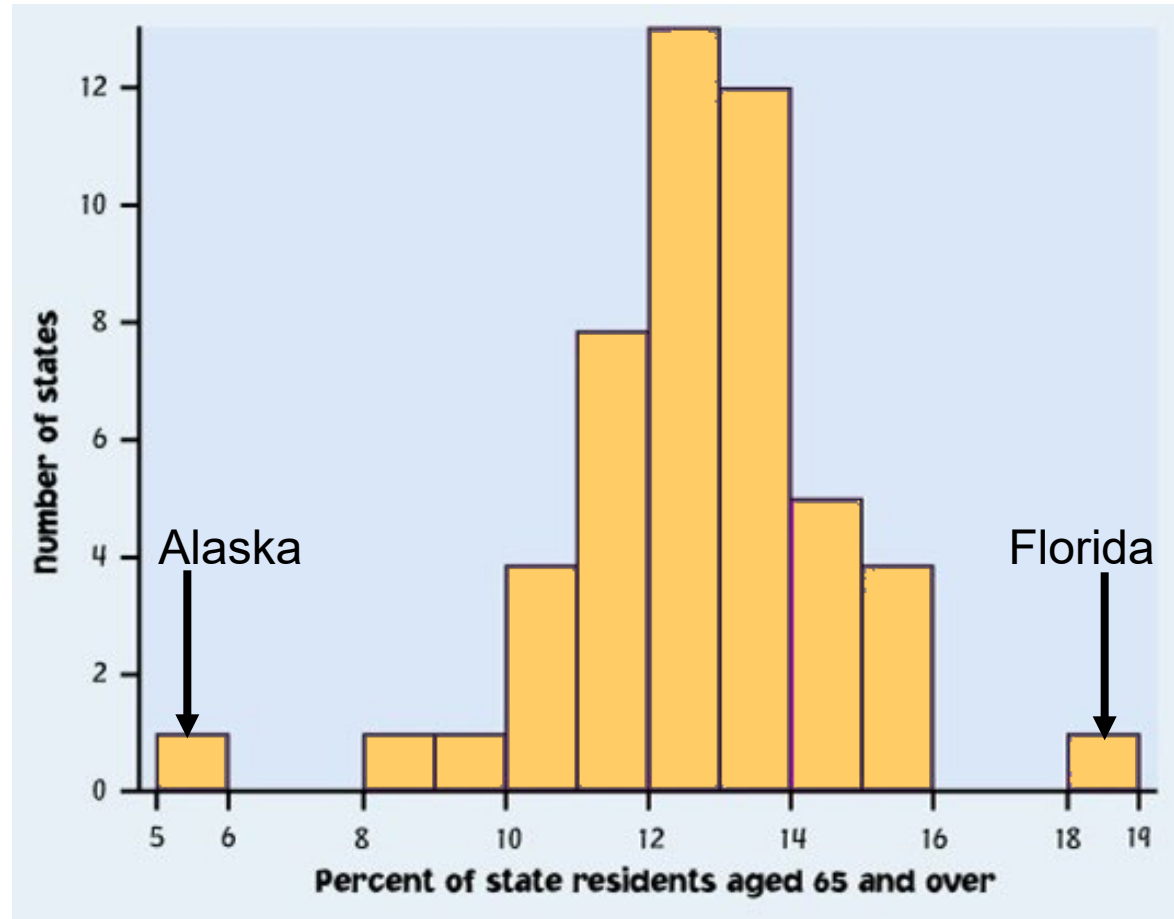


Outliers

An important kind of deviation is an **outlier**. Outliers are observations that lie outside the overall pattern of a distribution. Always look for outliers and try to explain them.

The overall pattern is fairly symmetrical except for two states that clearly do not belong to the main pattern. Alaska and Florida have unusually small and large percents, respectively, of elderly residents in their populations.

A large gap may indicate a possible outlier.

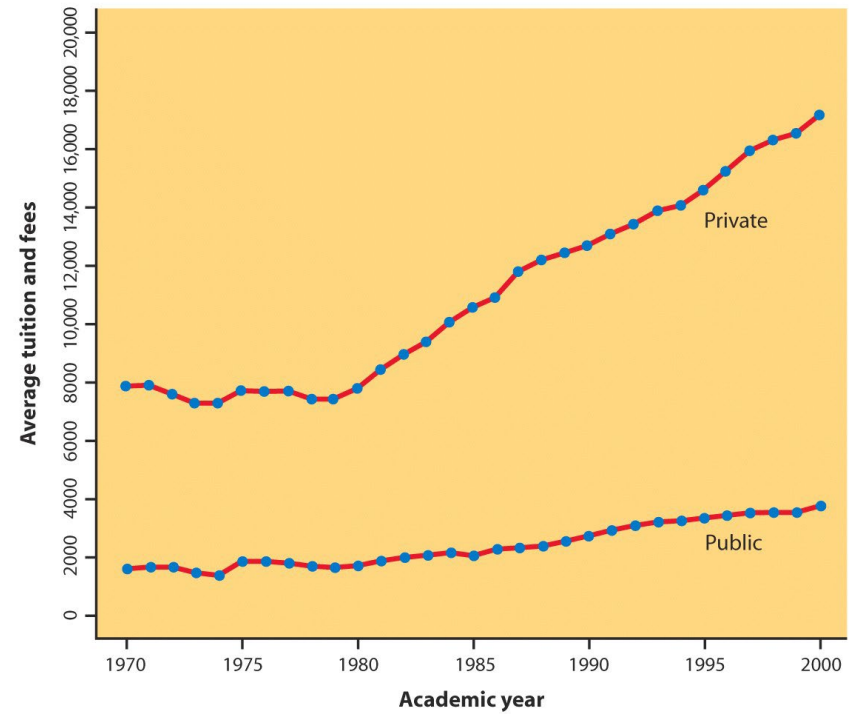
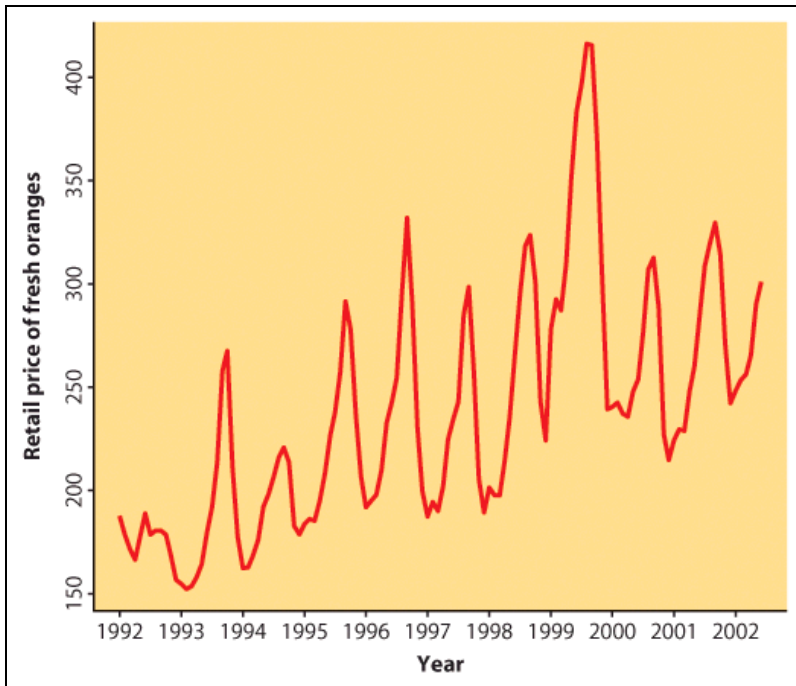


A **time plot** shows behavior over time.

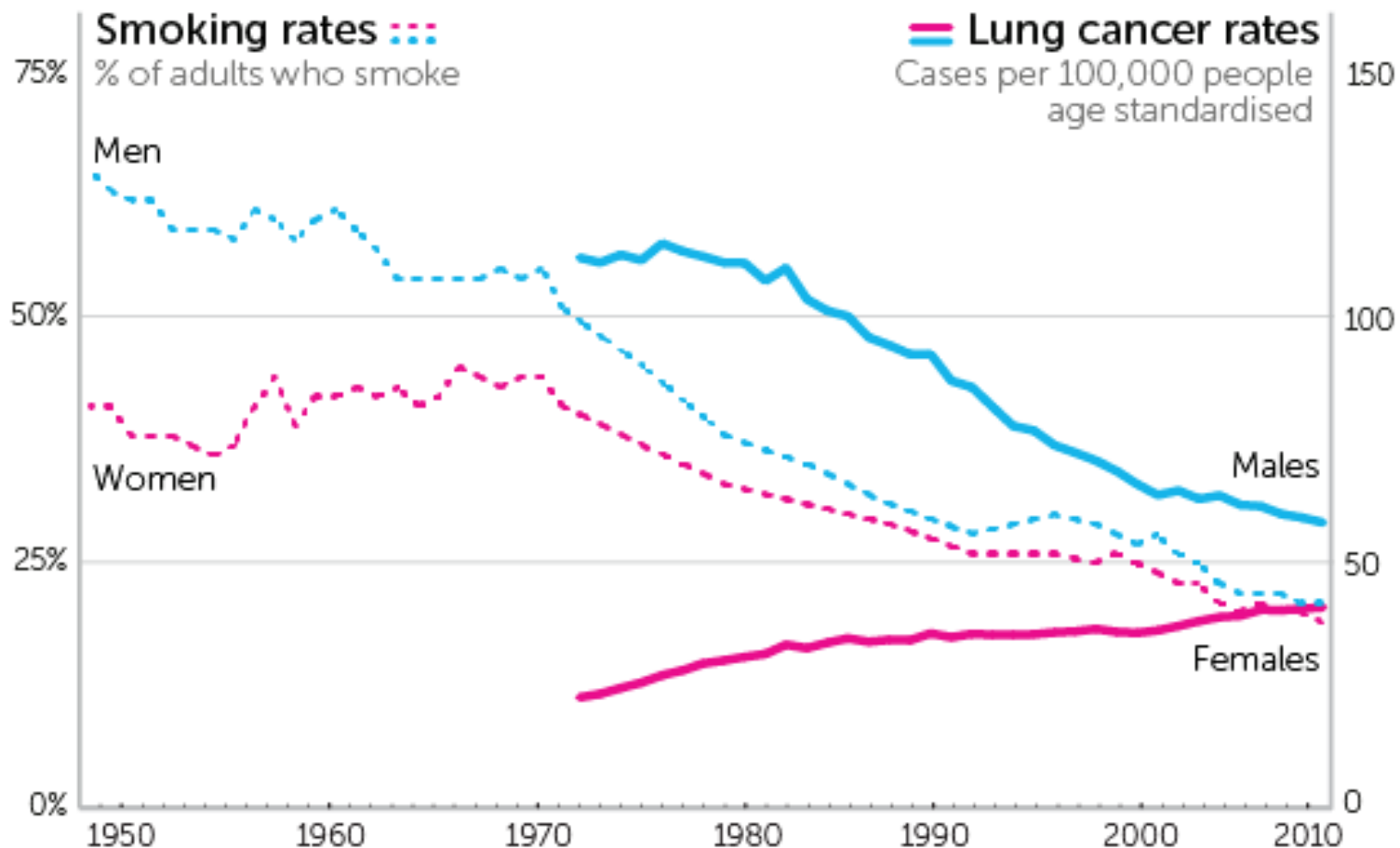
- Time is always on the horizontal axis, and the variable being measured is on the vertical axis.
- Look for an overall pattern (trend) and deviations from this trend. Connecting the data points by lines may emphasize this trend.
- Look for patterns that repeat at known regular intervals (seasonal variations) – COVID-19 confirmed cases

Time Plots 2

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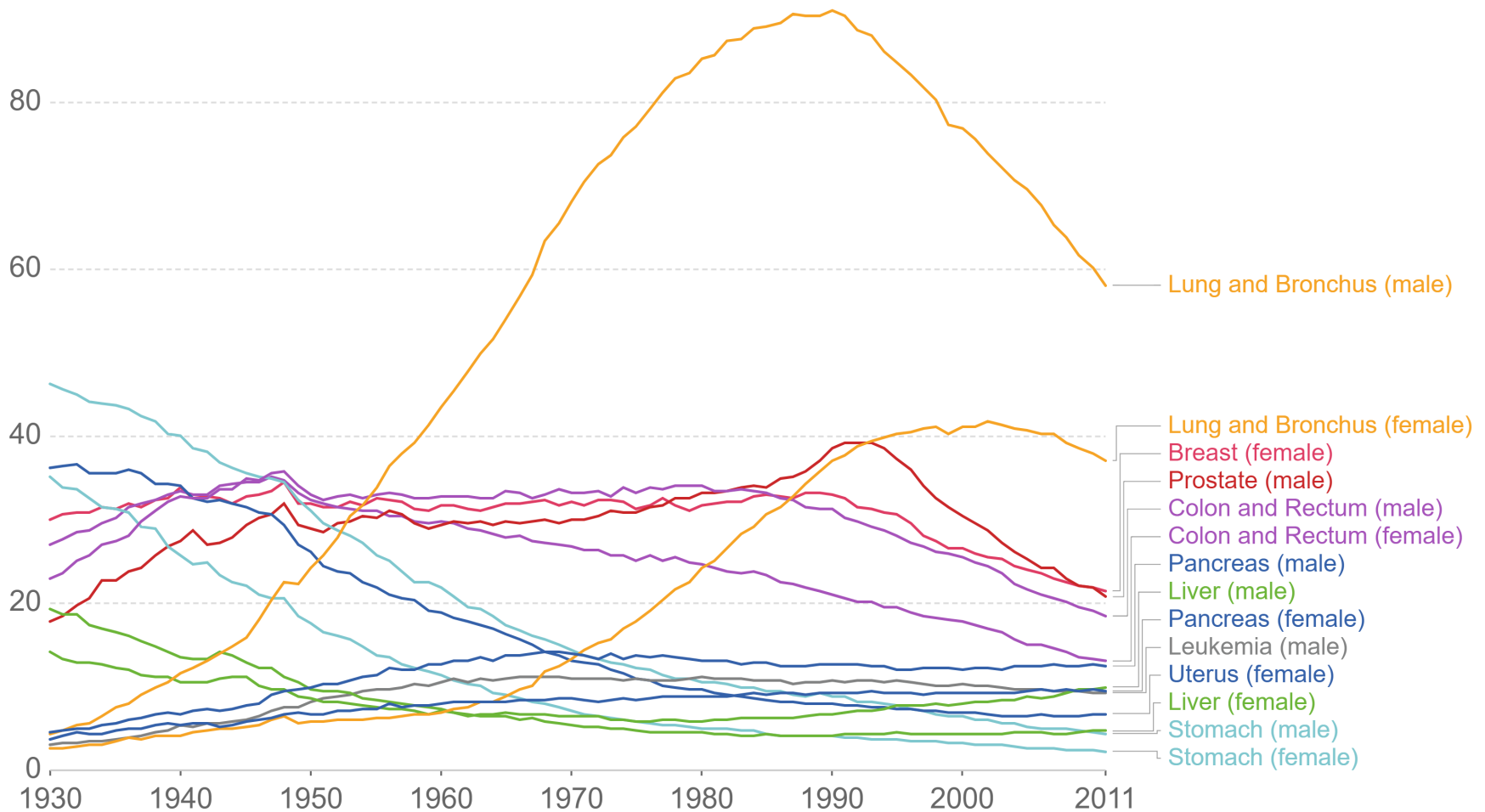


SMOKING RATES AND LUNG CANCER RATES IN THE UK



Cancer death rates in the United States over the long-run

Age-standardized death rates from various forms of cancer in males and females, measured as the number of deaths per 100,000 individuals. Age-standardization is based on normalisation to the standard US population structure in the year 2000.



Source: American Cancer Society

CC BY

1.3 Describing Distributions with Numbers

- Measuring center: mean
- Measuring center: median
- Measuring spread: quartiles
- Five-number summary and boxplot
- Measuring spread: standard deviation
- Choosing among summary statistics
- Changing the unit of measurement

Measuring Center: The Mean

The most common measure of center is the arithmetic average, or **mean**.

To find the **mean** $\bar{x}$ (pronounced “x-bar”) of a set of observations, add their values and divide by the number of observations. If the n observations are $x_1, x_2, x_3, \dots, x_n$, their mean is

$$\bar{x} = \frac{\text{sum of observations}}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

In a more compact notation:

$$\bar{x} = \frac{1}{n} \sum x_i$$

Measuring Center: The Median

Because the mean cannot resist the influence of extreme observations, it is not a **resistant** or **robust** measure of center.

Another common measure of center is the **median**.

The **median M** is the midpoint of a distribution, the number such that half of the observations are smaller and the other half are larger.

To find the median of a distribution:

1. Arrange all observations from smallest to largest.
2. If the number of observations n is odd, the median M is the center observation in the ordered list.
3. If the number of observations n is even, the median M is the average of the two center observations in the ordered list.

Measuring Center: Example

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Use the data below to calculate the mean and median of the time to start a business (in days) of 24 randomly selected countries.

16	4	5	6	5	7	12	19	10	2	25	19
38	5	24	8	6	5	53	32	13	49	11	17

$$\bar{x} = \frac{16 + 4 + 5 + 6 + \dots + 11 + 17}{24} = 16.292 \text{ days}$$

0	2455556678
1	01236799
2	45
3	28
4	9
5	3

Key: 4|9 represents
a country that had a
49-day time to start
a business.

$$M = \frac{11 + 12}{2} = 11.5 \text{ days}$$

Comparing Mean and Median

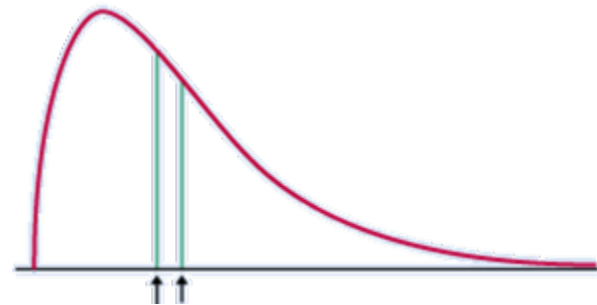
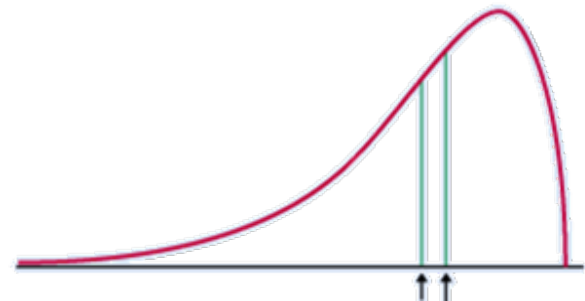
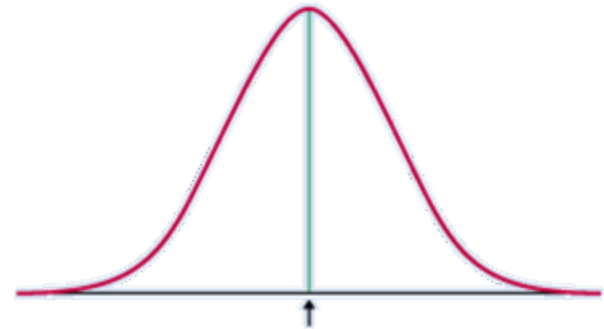
35

The mean and median measure center in different ways, and both are useful.

The mean and median of a roughly **symmetric** distribution are close together.

If the distribution is exactly **symmetric**, the mean and median are exactly the same.

In a **skewed** distribution, **the mean is usually farther out in the long tail than is the median.**



Measuring Spread: The Quartiles

- A measure of center alone can be misleading.
- A useful numerical description of a distribution requires both a measure of center and a measure of spread.

Calculating the Quartiles

- Arrange the observations in increasing order and locate the **median M** .
- The **first quartile Q_1** is the median of the observations located to the *left* of the median in the ordered list.
- The **third quartile Q_3** is the median of the observations located to the *right* of the median in the ordered list.

The Five-Number Summary

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The minimum and maximum values alone tell us little about **the distribution as a whole**. Likewise, the median and quartiles tell us little about **the tails** of a distribution.

To get a quick summary of both center and spread, combine all five numbers.

The **five-number summary** of a distribution consists of the smallest observation, the first quartile, the median, the third quartile, and the largest observation, written in order from smallest to largest.

Minimum Q_1 M Q_3 Maximum

The median and quartiles divide the distribution roughly into quarters. This leads to a new way to display quantitative data, the **boxplot**.

How to Make a Boxplot

- Draw and label a number line that includes the range of the distribution.
- Draw a central box from Q_1 to Q_3 .
- Note the median M inside the box.
- Extend lines (whiskers) from the box out to the minimum and maximum values that **are not outliers (defined via the interquartile range (IQR))**.

Suspected Outliers: $1.5 \times \text{IQR}$ Rule

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The **interquartile range (IQR)** is defined as $\text{IQR} = Q_3 - Q_1$.

In addition to serving as a measure of spread, the IQR is used as part of a rule of thumb for identifying outliers.

The $1.5 \times \text{IQR}$ Rule for Outliers

Call an observation an outlier if it falls more than $1.5 \times \text{IQR}$ above the third quartile or below the first quartile.

In the business start time data, $Q_1 = 5.5$ days, $Q_3 = 21.5$ days, and so $\text{IQR} = 16$ days.

For these data, $1.5 \times \text{IQR} = 1.5(16) = 24$

$Q_1 - 1.5 \times \text{IQR} = 5.5 - 24 = -18.5$

$Q_3 + 1.5 \times \text{IQR} = 21.5 + 24 = 45.5$

Any business start time shorter than -18.5 days or longer than 45.5 days is considered an outlier.

0	2455556678
1	01236799
2	45
3	28
4	9
5	3

Boxplots 2

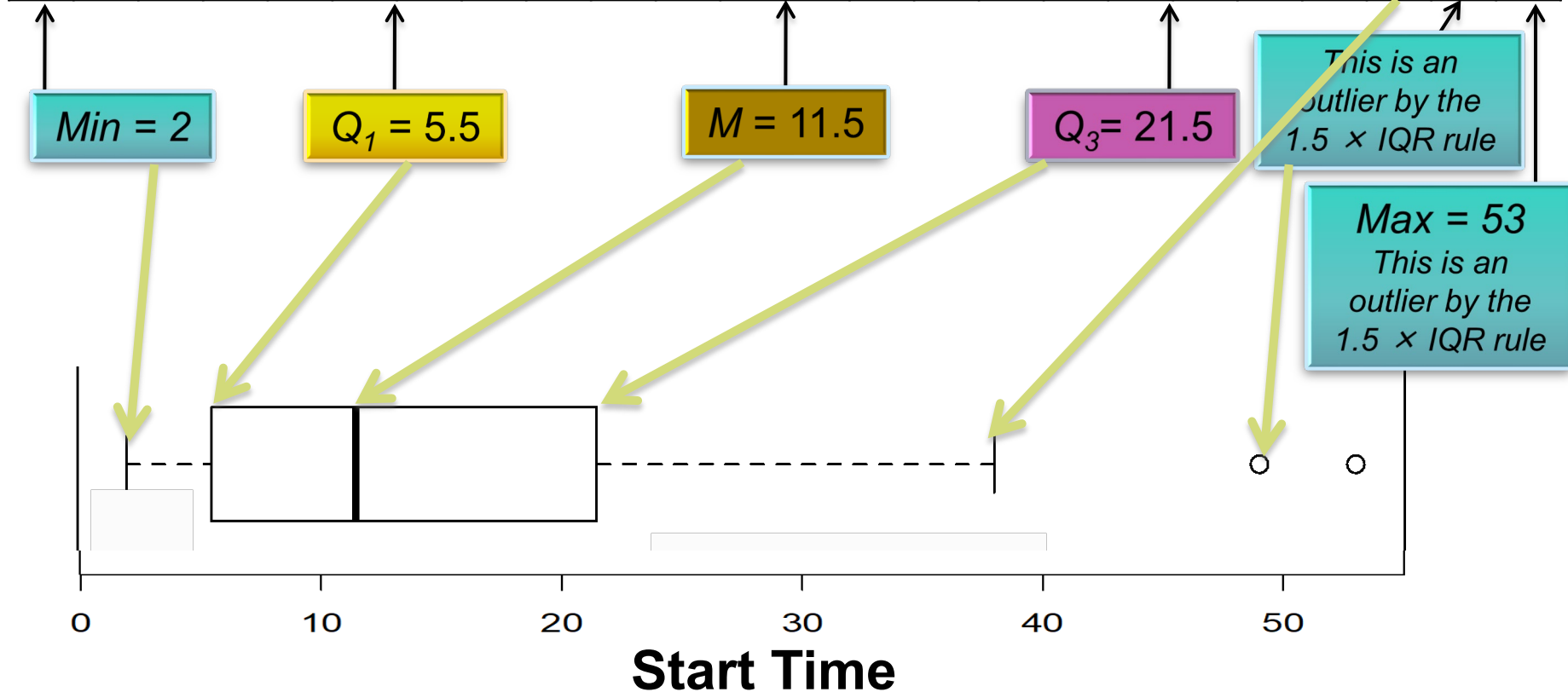
40

Consider our business start times data. Construct a boxplot.

16	4	5	6	5	7	12	19	10	2	25	19
38	5	24	8	6	5	53	32	13	49	11	17

Sort the data

2	4	5	5	5	5	6	6	7	8	10	11	12	13	16	17	19	19	24	25	32	38	49	53
---	---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----	----	----	----



Measuring Spread: The Standard Deviation

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The most common measure of spread looks at *how far each observation is from the mean*. This measure is called the **standard deviation**.

The (**sample**) **standard deviation** s_x measures the *average distance of the observations from their mean*. It is calculated by finding an average of the squared distances and then taking the square root. This average squared distance is called the **variance**.

$$\text{variance} = s_x^2 = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n - 1} = \frac{1}{n - 1} \sum (x_i - \bar{x})^2$$

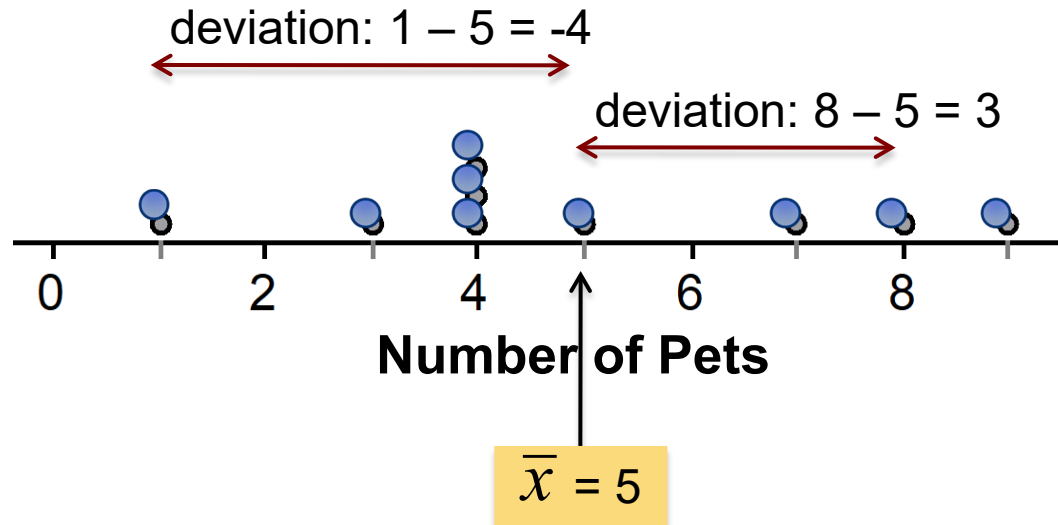
$$\text{standard deviation} = s_x = \sqrt{\frac{1}{n - 1} \sum (x_i - \bar{x})^2}$$

Calculating the Standard Deviation 1

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Example: Consider the following data on the number of pets owned by a group of nine children.

1. Calculate the mean.
2. Calculate each *deviation*.
deviation = *observation* – *mean*



Calculating the Standard Deviation 2

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3. Square each deviation.
4. Find the “average” squared deviation. Calculate the sum of the squared deviations divided by $(n - 1)$. This is called the **variance**.
5. Calculate the square root of the variance. This is the **standard deviation**.

x_i	$(x_i - \text{mean})$	$(x_i - \text{mean})^2$
1	$1 - 5 = -4$	$(-4)^2 = 16$
3	$3 - 5 = -2$	$(-2)^2 = 4$
4	$4 - 5 = -1$	$(-1)^2 = 1$
4	$4 - 5 = -1$	$(-1)^2 = 1$
4	$4 - 5 = -1$	$(-1)^2 = 1$
5	$5 - 5 = 0$	$(0)^2 = 0$
7	$7 - 5 = 2$	$(2)^2 = 4$
8	$8 - 5 = 3$	$(3)^2 = 9$
9	$9 - 5 = 4$	$(4)^2 = 16$
	Sum = ?	Sum = ?

“Average” squared deviation = $52 / (9 - 1) = 6.5$. This is the **variance**.

Standard deviation = square root of variance = $\sqrt{6.5} = 2.55$

Properties of the Standard Deviation

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- s measures spread about the mean and should be used only when the mean is an appropriate measure of center.
- $s = 0$ only when all observations have the same value and there is no spread. Otherwise, $s > 0$.
- s is *not* resistant to outliers.
- s has the **same units of measurement** as the original observations.

Choosing Measures of Center and Spread

We now have a choice between two descriptions for center and spread:

- ✓ Mean and standard deviation
- ✓ Median and interquartile range

Choosing Measures of Center and Spread

The median and *IQR* are usually better than the mean and standard deviation for describing a *skewed distribution* or a distribution *with outliers*.

Use mean and standard deviation only for reasonably symmetric distributions that do not have outliers.

NOTE: Numerical summaries do not fully describe the *shape* of a distribution. *ALWAYS PLOT YOUR DATA!* The numerical summaries and graphical displays complement one another.

Changing the Unit of Measurement

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Variables can be recorded in different units of measurement. Most often, one measurement unit is a **linear transformation** of another measurement unit: $x_{\text{new}} = a + bx$.

Linear transformations do not change the **basic shape** of a distribution (skew, symmetry, multimodal). But **they do change the measures of center and spread**.

- Multiplying each observation by a positive number b multiplies both measures of center (mean, median) and spread (IQR, s) by b .
- Adding the same number a (positive or negative) to each observation adds a to measures of center and to quartiles, but it does not change measures of spread (IQR, s).

1.4 Density Curves and Normal Distributions

- Density curves
- Measuring center and spread for density curves
- Normal distributions
- The 68-95-99.7 rule
- Standardizing observations
- Using the standard normal table
- Inverse normal calculations
- Normal quantile plots

Exploring Quantitative Data

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We now have a kit of graphical and numerical tools for describing distributions. We also have a strategy for exploring data on a single quantitative variable. Now we'll add a **fourth** step to the strategy.

Exploring Quantitative Data

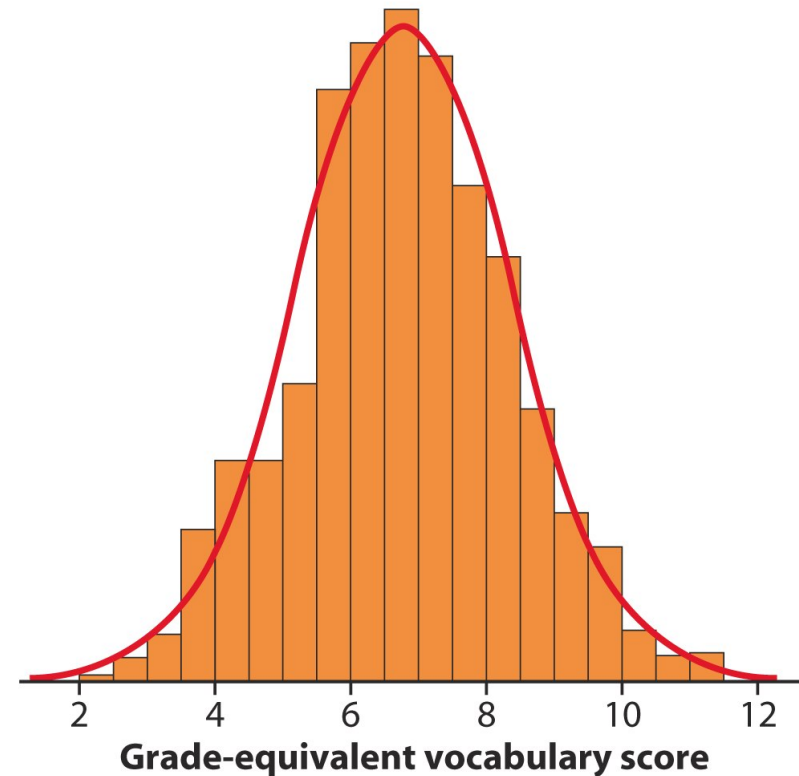
1. Always plot your data: make a graph.
2. Look for the overall pattern (shape, center, and spread) and for striking departures such as outliers.
3. Calculate a numerical summary to briefly describe center and spread.
4. Sometimes, the overall pattern of *a large number* of observations is so regular that we can describe it by a *smooth curve*.

Density Curves 1

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Example: Here is a histogram of vocabulary scores of 947 seventh graders.

The smooth curve drawn over the histogram is a **mathematical model** for the distribution.

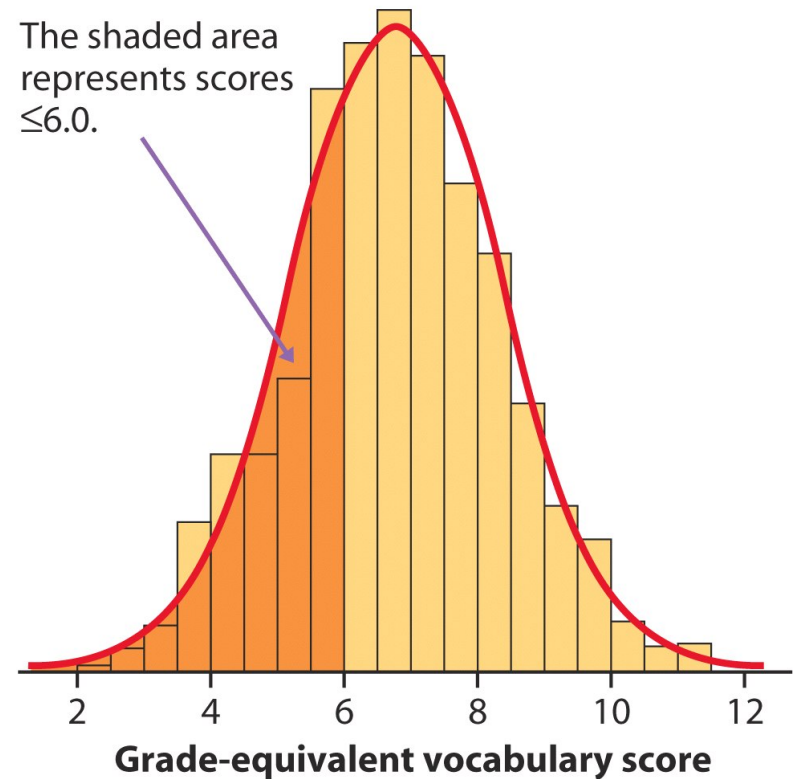


Density Curves 2

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The areas of the **shaded bars** in this histogram represent the **proportion of scores** in the observed data that are less than or equal to 6.0. This proportion is equal to **0.303**.

Now the area under the smooth curve to the left of 6.0 is shaded. *If the scale is adjusted so the total area under the curve is exactly 1*, then this curve is called a **density curve**. The **proportion of the area** to the left of 6.0 is now equal to **0.293**.



Density Curves 3

A **density curve** is a curve that

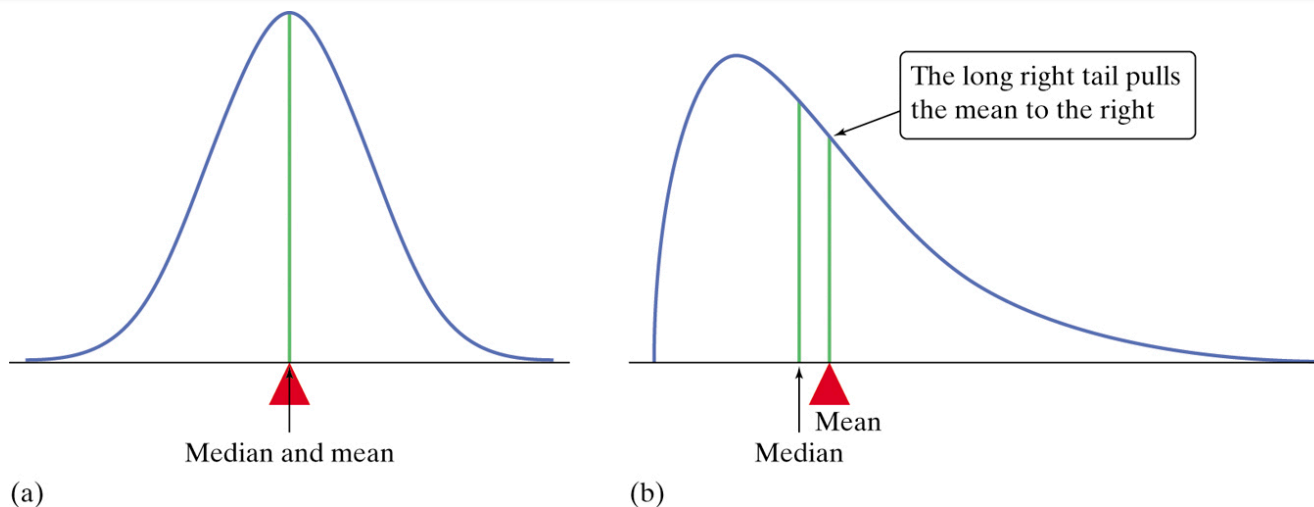
- is always **on or above** the horizontal axis.
- has an area of **exactly 1** underneath it.

A density curve describes the overall pattern of a distribution. The area under the curve and above any range of values on the horizontal axis is the proportion of all observations that fall in that range.

Our measures of center and spread apply to density curves as well as to actual sets of observations.

Distinguishing the Median and Mean of a Density Curve

- The **median** of a density curve is the “equal-areas” point—the point that divides the area under the curve in half.
- The **mean** of a density curve is the balance point, that is, **the point at which the curve would balance if made of solid material**.
- The median and the mean are the same for a symmetric density curve. They both lie at the center of the curve. The mean of a skewed curve is pulled away from the median in the direction of the long tail.



Density Curves 5

- The mean and standard deviation computed from **actual observations** (data) are denoted by $\bar{x}$ and s , respectively.
- The mean and standard deviation of the actual distribution (**likely not observed**) represented by the **density curve** are denoted by μ (“mu”) and σ (“sigma”), respectively.
- (Statistics vs Parameters)

Normal Distributions 1

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One particularly important class of density curves is the class of normal curves, which describe Normal distributions.

- All normal curves are symmetric, single-peaked, and bell-shaped.
- A specific normal curve is described by giving its mean μ and standard deviation σ .

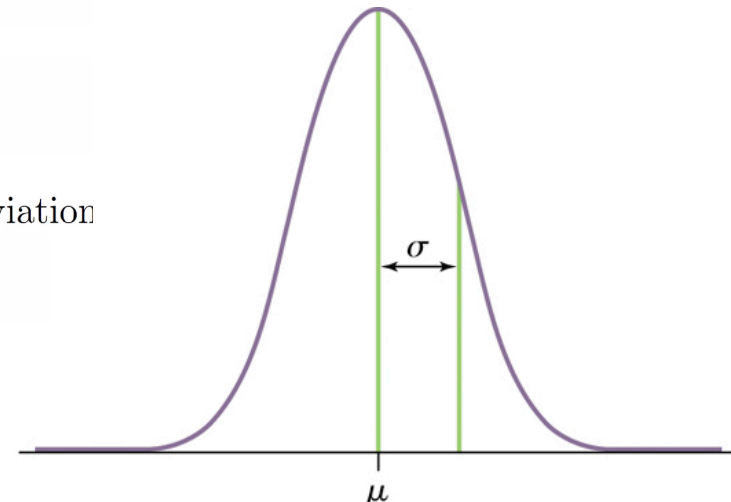
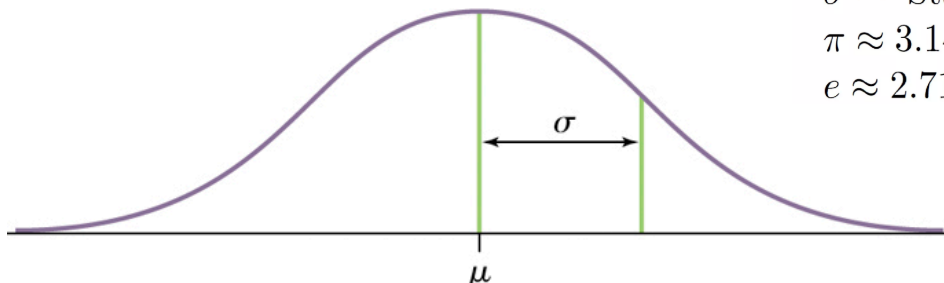
$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

μ = Mean

σ = Standard Deviation

$\pi \approx 3.14159 \dots$

$e \approx 2.71828 \dots$



A **normal distribution** is described by a normal density curve. Any particular normal distribution is **completely specified** by two numbers: its mean μ and standard deviation σ .

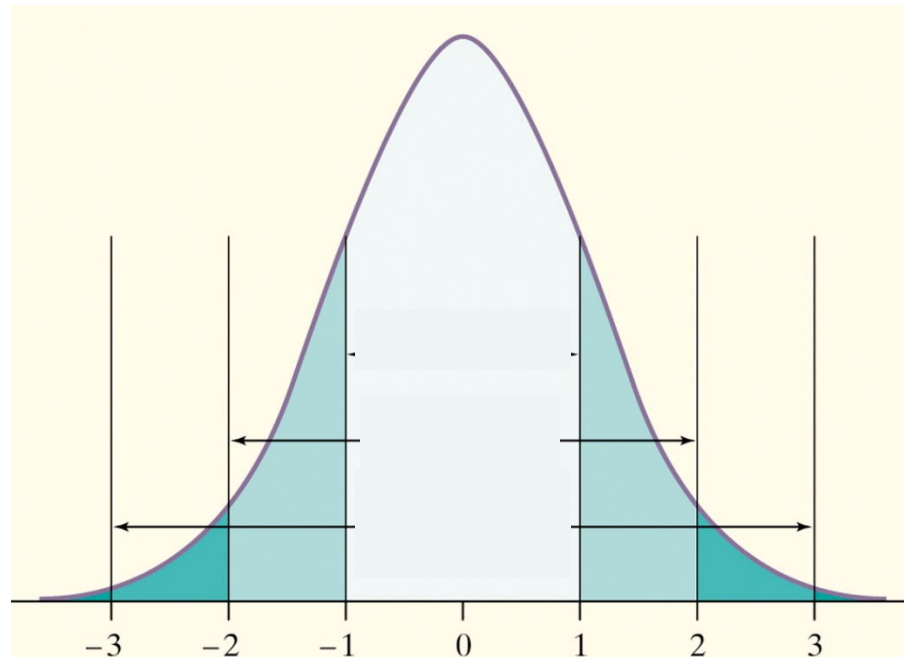
- The mean of a normal distribution is the center of the symmetric **normal curve**.
- The standard deviation is the distance from the center to the change-of-curvature points on either side.
- We abbreviate the normal distribution with mean μ and standard deviation σ as **$N(\mu, \sigma)$** .

The 68-95-99.7 Rule 1

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In the normal distribution with mean μ and standard deviation σ :

- approximately **68%** of the observations fall within σ of μ .
- approximately **95%** of the observations fall within 2σ of μ .
- approximately **99.7%** of the observations fall within 3σ of μ .

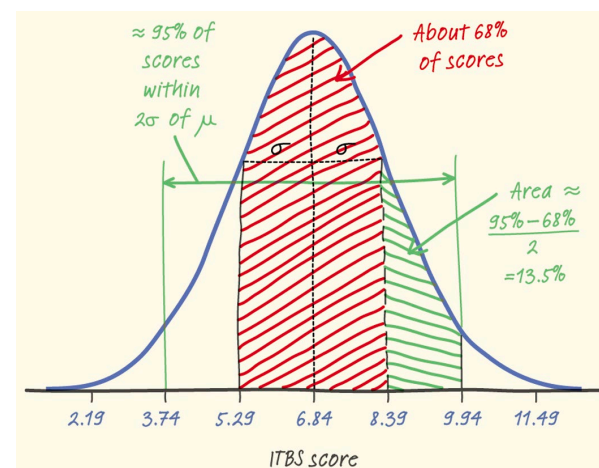
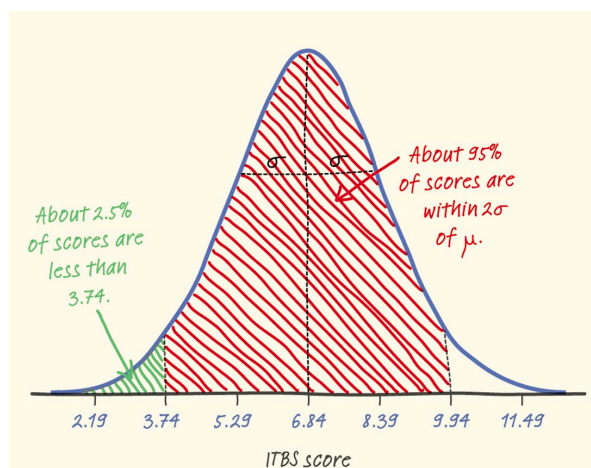
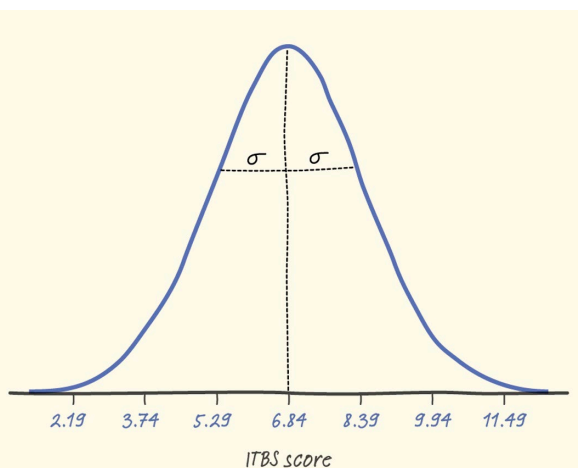


The 68-95-99.7 Rule 2

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The distribution of Iowa Test of Basic Skills (ITBS) vocabulary scores for 7th grade students in Gary, Indiana, is close to normal. Suppose the distribution is $N(6.84, 1.55)$.

- ✓ Sketch the normal density curve for this distribution.
- ✓ What percent of ITBS vocabulary scores is less than 3.74?
- ✓ What percent of the scores is between 5.29 and 9.94?



Standardizing Observations 2

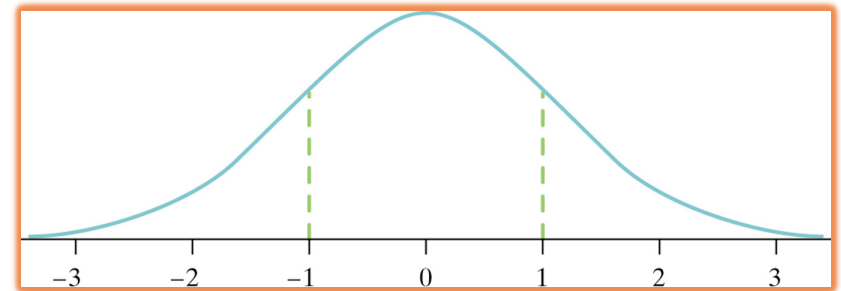
58

If a variable x has a distribution with mean μ and standard deviation σ , then the **standardized value** of x , or its **z-score**, is

$$z = \frac{x - \mu}{\sigma}$$

After standardization, all normal distributions can be expressed on the same scale (i.e., the standard normal distribution), with mean 0 and standard deviation 1.

The **standard normal distribution** is the normal distribution with mean 0 and standard deviation 1. That is, the standard normal distribution is $N(0,1)$.



The Standard Normal Table 1

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Because all normal distributions are the same when we standardize, we can find areas under any normal curve from a single table.

The Standard Normal Table

Table A is a table of areas under the standard normal curve. The table entry for each value z is the area under the curve **to the left** of z .

TABLE A Standard normal probabilities

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681

The Standard Normal Table 2

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Suppose we want to find the proportion of observations from the standard normal distribution that are less than 0.81

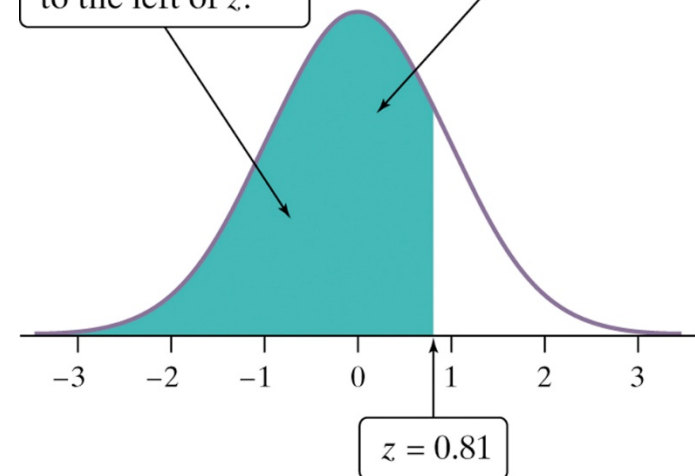
We can use Table A.

Z	.00	.01	.02
0.7	.7580	.7611	.7642
0.8	.7881	.7910	.7939
0.9	.8159	.8186	.8212

$$P(z < 0.81) = .7910$$

Table entry for z is always the area under the curve to the left of z .

Table entry = 0.7910 for $z = 0.81$.

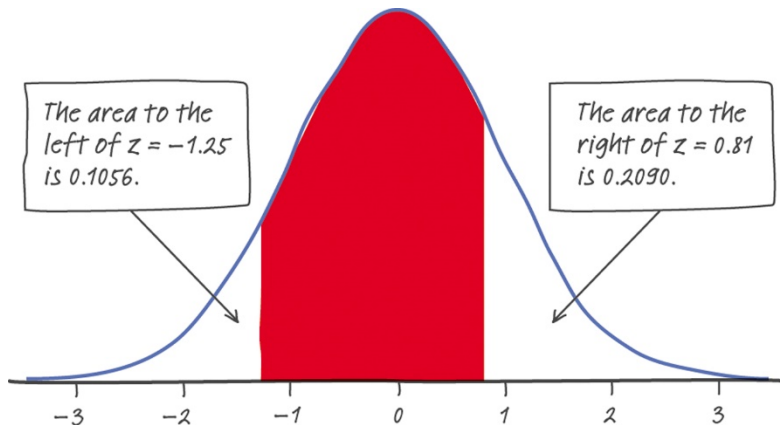


Normal Calculations 1

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Find the proportion of observations from the standard normal distribution that are between -1.25 and 0.81 .

Can you find the same proportion using a different approach?



$$1 - (0.1056 + 0.2090) = 1 - 0.3146$$
$$= \mathbf{0.6854}$$

How to Solve Problems Involving Normal Distributions

Express the problem in terms of the observed variable x .

Draw a picture of the distribution and shade the area of interest under the curve.

Perform calculations.

- **Standardize** x to restate the problem in terms of a standard Normal variable z .
- **Use Table A** and the fact that the total area under the curve is 1 to find the required area under the standard normal curve.

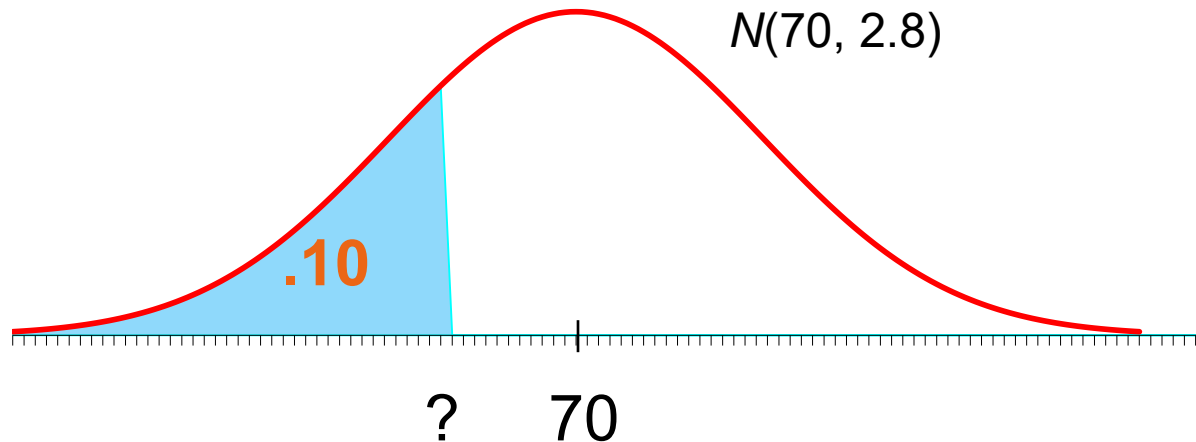
Write your conclusion in the context of the problem.

Normal Calculations 3

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According to the National Health and Nutrition Examination Study (NHANES) of 1976–1980, the heights (in inches) of men aged 18–24 are $N(70, 2.8)$.

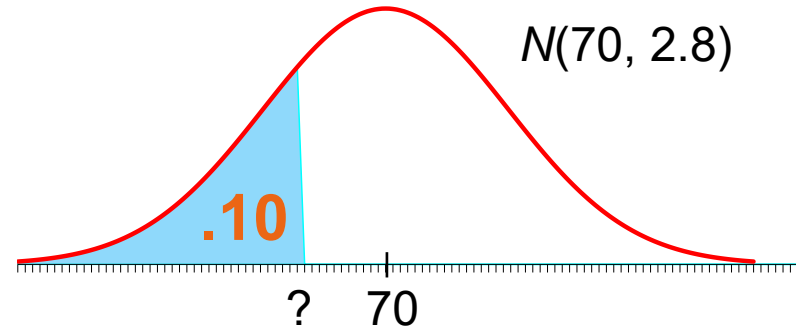
If exactly 10% of men aged 18–24 are shorter than a particular man, how tall is he?



Normal Calculations 4

64

If exactly 10% of men aged 18–24 are shorter than a particular man, how tall is he?



Look up the probability closest to 0.10 in the table.

Find the corresponding **standardized score**.

The value you seek is **that many standard deviations** from the mean.

z	.07	.08	.09
-1.3	.0853	.0823	.0823
-1.2	.1003	.0985	.0985
-1.1	.1210	.1190	.1170

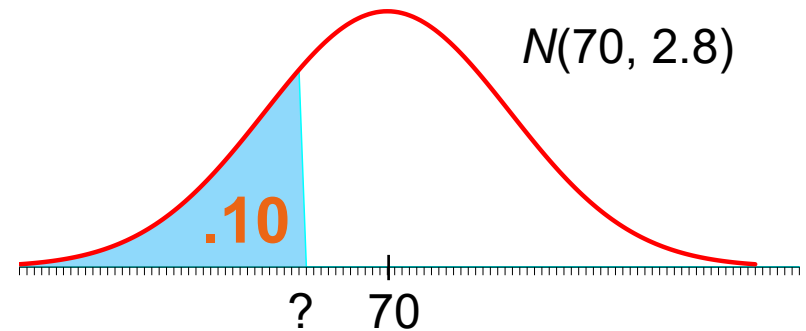
$$Z = -1.28$$

Normal Calculations 5

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If exactly 10% of men aged 18–24 are shorter than a particular man, how tall is he?

$$Z = -1.28$$



We need to “unstandardize” the z-score to find the observed value (x):

$$Z = \frac{x - \mu}{\sigma} \Rightarrow x = \mu + z\sigma$$

$$\begin{aligned} x &= 70 + z(2.8) \\ &= 70 + [(-1.28) \times (2.8)] \\ &= 70 + (-3.58) = \underline{\underline{66.42}} \end{aligned}$$

A man would have to be approximately 66.42 inches tall or less to be in the lower 10% of all men in the population.

Normal Quantile Plots 1

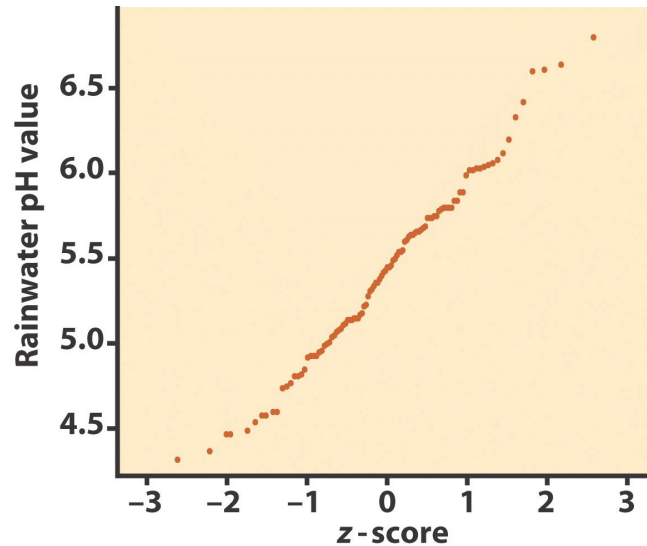
One way to assess if a distribution is indeed approximately Normal is to plot the data on a **normal quantile plot**.

The data points are *ranked* and the *percentile ranks are converted to z-scores* with Table A. The z-scores are then used for the x axis against which the data are plotted on the y axis of the normal quantile plot.

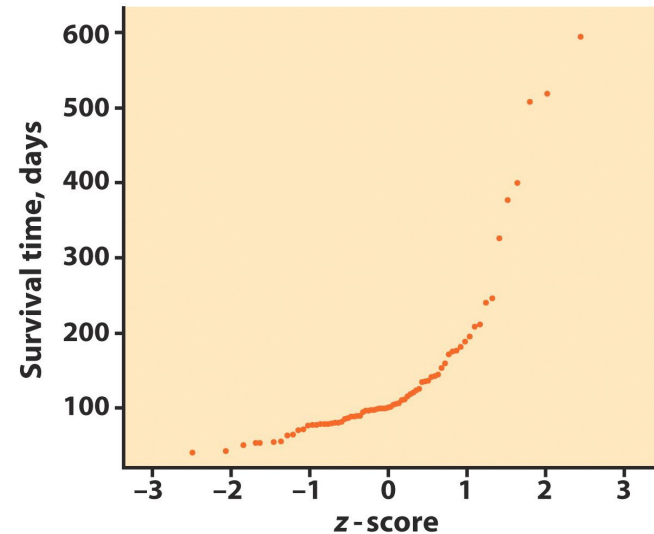
- If the distribution is indeed normal, the plot will show a *straight line*, indicating a good match between the data and a normal distribution.
- Systematic deviations from a straight line indicate a non-normal distribution. *Outliers appear as points that are far away from the overall pattern of the plot.*

Normal Quantile Plots 2

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Good fit to a straight line: The distribution of rainwater pH values is close to normal.



Curved pattern: The data are not normally distributed. Instead, the data are *right skewed*: A few individuals have particularly long survival times.

Normal quantile plots are complex to do by hand, but they are standard features in most statistical software.

Chapter 1

Looking at Data— Distributions (Review)

Introduction

1.1 Data

1.2 Displaying Distributions with Graphs

1.3 Describing Distributions with Numbers

1.4 Density Curves and Normal Distributions